

Projective Space

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1 Proj Construction

Let S be a $(\mathbb{Z}^{\geq 0})$ -graded ring.

Remark 1.1. If $T \subseteq S$ is a multiplicative set consisting of homogeneous elements, then $T^{-1}S$ is naturally a $\mathbb{Z}$ -graded ring, where we set

$$\deg \frac{g}{t} = \deg g - \deg t.$$

In analogy to the classical setting, where $S = \mathbb{C}[x_0, \dots, x_n]/I$ for I a homogeneous ideal, we will define $\text{Proj } S$ by gluing together the dehomogenizations of S along coordinates. To generalize, however, we allow *any* coordinate, not just the “standard” coordinates x_i .

Let’s recall how dehomogenization works classically. For example, for example, consider $S = \mathbb{C}[x_0, x_1, x_2]/(g)$, where

$$g(x_0, x_1, x_2) = x_0 x_1 + x_2^2.$$

Classically, we pick a coordinate, say x_0 , and look at the defining equation of g in the complement of its vanishing $U_0 = \mathbb{P}^3 \setminus V(x_0)$. We would realize that in homogeneous coordinates we can rescale the nonzero x_0 to be 1, which is effectively dividing the coordinates by x_0 . So, we would evaluate

$$g\left(\frac{x_0}{x_0}, \frac{x_1}{x_0}, \frac{x_2}{x_0}\right) = \frac{x_0}{x_0} \frac{x_1}{x_0} + \left(\frac{x_2}{x_0}\right)^2,$$

and we might rename these new variables $x_{1/0} = \frac{x_1}{x_0}$, $x_{2/0} = \frac{x_2}{x_0}$. Then, our chart with respect to x_0 is

$$\text{Spec } \mathbb{C}[x_{1/0}, x_{2/0}]/(x_{1/0} + x_{2/0}^2).$$

This coordinate ring is exactly the degree 0 part of the ring $(\mathbb{C}[x_0, x_1, x_2]/g)_{x_0}$.

Now, how should this work more generally? We want a notion of dehomogenizing at an arbitrary homogeneous $f \in S_+$, which we interpret as a coordinate. Now, we again study the complement of the vanishing of the coordinate, which is captured algebraically by S_f . Then, we look at the degree 0 part—this is because we want a standard representative of functions which are all equivalent. So, we will declare

$$\text{Spec}(S_f)_0$$

to be a chart of $\text{Proj } S$ (which is still yet to be defined).

As a set, define

$$\text{Proj } S = \{\text{homogeneous prime ideals of } S \text{ not containing } S_+\}.$$

Proposition 1.2. Let A be a $\mathbb{Z}$ -graded ring with an invertible positive degree element f (e.g., $A = S_f$). Then, there is a bijection between the prime ideals of A_0 and the homogeneous prime ideals of A_f by

$$\{\text{homog. primes of } A\} \leftrightarrow \{\text{primes of } A_0\}$$

$$P \mapsto P \cap A_0$$

$$\bigoplus Q_i \hookleftarrow P_0$$

where $Q_i \subseteq A_i$ is the set

$$Q_i = \{a \in A_i \mid a^{\deg f} / f^i \in P_0\}.$$

Proof. The left-to-right direction gives a well-defined prime ideal as usual for a map of rings $A_0 \hookrightarrow A$.

We show that the right-to-left is well-defined, so we show

$$Q = \bigoplus Q_i$$

is a homogeneous prime ideal. Once we show it is an ideal, it is automatically homogeneous by its construction (generated by homogeneous forms in each Q_i).

First, we prove a lemma, that $a \in Q_i$ iff $a^2 \in Q_{2i}$. If $a \in Q_i$, then $a^{\deg f}/f^i \in P_0$, and since P_0 is an ideal, we can scale it to get

$$\frac{a^{\deg f}}{f^i} \cdot \frac{a^{\deg f}}{f^i} = \frac{(a^2)^{\deg f}}{f^{2i}} \in P_0,$$

so $a^2 \in Q_{2i}$. Conversely, if $a^2 \in Q_{2i}$, then P_0 being prime and the previous equation gives $a \in Q_i$.

Now, to show Q is at least an abelian subgroup, we just need to show each Q_i is an abelian subgroup, so let $a, b \in Q_i$. To show $a + b \in Q_i$, we show $(a + b)^2 \in Q_{2i}$. We have

$$\frac{((a + b)^2)^{\deg f}}{f^{2i}} = \frac{(a + b)^{2 \deg f}}{f^{2i}},$$

and when we distribute, by pigeonhole, each term will have at least one of a or b a power at least $\deg f$ (without loss of generality, say a), so

$$\frac{a^m b^n}{f^{2i}} = \frac{a^{m-\deg f} b^n}{f^i} \frac{a^{\deg f}}{f^i},$$

which is in P_0 because it is an ideal, so we get that $a + b \in Q_i$. Thus, Q_i is indeed an abelian group, and it is not hard to show that it is an ideal, so the right-to-left construction is well-defined.

Now, we show that this actually gives a bijection, i.e., that the directions are inverse to each other. For the composition

$$P_0 \mapsto \bigoplus Q_i \mapsto Q_0,$$

by definition $Q_0 = \{a \in A_0 \mid a^{\deg f} \in P_0\}$, which tells us

$$P_0 \subseteq Q_0 \subseteq \sqrt{P_0},$$

but P_0 is radical anyways, so $Q_0 = P_0$.

Conversely, for the composition

$$P \mapsto P \cap A_0 \mapsto \bigoplus Q_i,$$

we will at least have $\bigoplus Q_i \subseteq P$ since if $a^{\deg f}/f^i \in P \cap A_0$, then $a^{\deg f} \in P$, so by primeness $a \in P$. Then, for $a \in P$ is homogeneous, we have $a^{\deg f}/f^{\deg a} \in P \cap A_0$, so $a \in Q_{\deg a}$, which then implies $P \subseteq \bigoplus Q_i$.

Thus, these maps are indeed inverse, and the proposition holds. $\square$

This tells us that our definition of $\text{Proj } S$ as a set respects dehomogenization, and that we should/will be able to identify $\text{Spec}(S_f)_0$ with a subset of $\text{Proj } S$.

As a topological space, define the closed sets of $\text{Proj } S$ to be, for $I \leq S$ a homogeneous ideal,

$$V(I) = \{P \in \text{Proj } S \mid P \geq I\},$$

and one should also check that this makes $\text{Spec}(S_f)_0 \rightarrow \text{Proj } S$ an inclusion of an open set. In fact, defining

$$D(f) = \text{Proj } S \setminus V(f),$$

one should see that $\text{Spec}(S_f)_0$ factors through $D(f)$, and that this is a homeomorphism.

As a ringed space, define $\mathcal{O}_{\text{Proj } S}$ by gluing $\mathcal{O}_{D(f)} = \mathcal{O}_{\text{Spec}(S_f)_0}$ along intersections $D(fg)$ (checking the cocycle condition on $D(fgh)$).

1.1 Homogeneous Tilde

Let M be a $\mathbb{Z}$ -graded S -module. Define the quasicoherent sheaf $\widetilde{M}$ on $\text{Proj } S$ by defining it on the base of open sets $D(f)$ for $f \in S$ homogeneous. Set

$$\widetilde{M}(D(f)) = (M_f)_0.$$

In particular, a $\mathbb{Z}^{\geq 0}$ -graded ring S comes equipped with the graded module S^+ . When S is standard graded (i.e., generated in degree 1), we will denote

$$\mathcal{O}_{\text{Proj } S}(1) := \widetilde{S^+}.$$

Proposition 1.3. $\mathcal{O}_{\text{Proj } S}(1)$ is a line bundle.

Proof. We have to trivialize $\mathcal{O}_{\text{Proj } S}(1)$. Since S is generated in degree 1, $S^+ = \sum_{i \in I} S \cdot s_i$ for $s_i \in S_1$ and I some index set. This tells us that

$$\text{Proj } S = \bigcup_{i \in I} D(s_i),$$

since any homogeneous prime ideal not containing S^+ must miss some s_i . Then, on $D(s_i)$, the sheaf $\mathcal{O}_{\text{Proj } S}(1)$ has a nowhere vanishing section s_i , which gives an isomorphism

$$\mathcal{O}_{D(s_i)} \xrightarrow{\cdot s_i} \mathcal{O}_{\text{Proj } S}(1)|_{D(s_i)},$$

so indeed $\mathcal{O}_{\text{Proj } S}(1)$ is a line bundle. □

2 Morphisms to Projective Space

2.1 Some Basics

2.1.1 Using the Trivial Line Bundle

As a first attempt to give maps to projective space, let X be a k -scheme, and $f_0, \dots, f_r \in \mathcal{O}_X(X)$ some global functions that don't simultaneously vanish. Then, we want to define a map

$$X \xrightarrow{[f_0:\dots:f_r]} \mathbb{P}_k^r.$$

At least at k -points, we get a map of sets, where we send x to the (homogeneous maximal ideal corresponding to) k^* equivalence class $[f_0(x), \dots, f_r(x)]$, where $f_i(x) \in k$. To make more sense out of this map, we should say what happens to other points, and what happens to the structure sheaf. To do this, we locally define

$$D(f_i) \xrightarrow{(\frac{f_0}{f_i}, \dots, \frac{f_i}{f_i}, \dots, \frac{f_r}{f_i})} D(x_i) \subseteq \mathbb{P}_k^r,$$

where the $D(f_i)$ cover X because $V(f_0, \dots, f_r) = \emptyset$. Now, this morphism makes sense (on the level of schemes) by the Γ, Spec adjunction, where we are giving a ring homomorphism

$$\Gamma(D(x_i)) = (k[x_0, \dots, x_r]_{(x_i)})_0 \cong k[x_{0/i}, \dots, x_{i/i}, \dots, x_{r/i}] \longrightarrow \Gamma(D(f_i))$$

by sending $x_{j/i} \mapsto f_j/f_i \in \Gamma(D(f_i))$.

However, we don't have much hope of producing many morphisms in this way. For example, we cannot even produce the morphism $\mathbb{P}_k^r \xrightarrow{\text{id}} \mathbb{P}_k^r$, since $\mathbb{P}_k^r$ has only k for global sections. To have better luck, we need to use sections not just from $\mathcal{O}_X$, but from more complicated line bundles.

2.1.2 Basepoint-free Line Bundles

First, we discuss some adjectives about sheaves and their sections. Let X be a k -scheme and $\mathcal{F}$ a sheaf. We say $V \subseteq H^0(X, \mathcal{F})$ **generates** $\mathcal{F}$ if

$$V \otimes_k \mathcal{O}_X \xrightarrow{\text{eval}} \mathcal{F}$$

is surjective. If some V generates $\mathcal{F}$ (in particular $V = H^0(X, \mathcal{F})$ all global sections), we say $\mathcal{F}$ is **globally generated**.

In a similar vein, we say $\mathcal{F}$ is **globally generated at P** by V if evaluation of V at the stalk of P is surjective, i.e.,

$$V \otimes_k \mathcal{O}_{X,P} \twoheadrightarrow \mathcal{F}_P.$$

Proposition 2.1. Let $\mathcal{F}$ be a finite type (i.e., sections over affine open is finite generated) quasicohherent sheaf. Then, $\mathcal{F}$ is globally generated at P by V iff global sections generate the fibers at P by V , i.e.,

$$V \otimes_k \mathcal{O}_{X,P} / \mathfrak{m}_P \mathcal{O}_{X,P} \rightarrow \mathcal{F}_P / \mathfrak{m}_P \mathcal{F}_P$$

is surjective.

Proof. If $V \otimes_k \mathcal{O}_{X,P} \twoheadrightarrow \mathcal{F}_P$, then restrictions of sections in V to $\mathcal{F}_P$ generate the stalk, so their images generate the quotient $\mathcal{F}_P/\mathfrak{m}_P \mathcal{F}_P$. Alternatively, we could view this as tensoring by the skyscraper sheaf $k(P)$ is right exact, so preserves surjections.

Conversely, if $V \otimes_k \mathcal{O}_{X,P}/\mathfrak{m}_P \mathcal{O}_{X,P} \twoheadrightarrow \mathcal{F}_P/\mathfrak{m}_P \mathcal{F}_P$, then there are global sections $m_1, \dots, m_n \in V$ of $\mathcal{F}$ which restrict/quotient to generate the vector space $\mathcal{F}_P/\mathfrak{m}_P \mathcal{F}_P$, the residue field of a finitely generated $\mathcal{O}_P$ -module $\mathcal{F}_P$. So, by Nakayama's lemma, $m_1, \dots, m_n$ generate $\mathcal{F}_P$. $\square$

Proposition 2.2. If $\mathcal{F}$ is finite type and is globally generated at P by V , then there is a neighborhood U of P where $\mathcal{F}$ is globally generated at all points of U .

Proof. Since $\mathcal{F}$ is finite type and is globally generated at P by V , there are sections $s_1, \dots, s_r \in V$ with $\langle s_1, \dots, s_r \rangle = \mathcal{F}_P$. We want to show that $s_1, \dots, s_r$ generate the fibers for nearby Q (using the previous proposition).

In an affine open $\text{Spec } A \subseteq X$, $\mathcal{F} = \widetilde{M}$ for a finitely generated A -module. We have M_P generated by the images of elements $m_1, \dots, m_r \in M$ associated to $s_1, \dots, s_r$, and we want to show that they generate nearby M_Q . $\square$

In particular, if $\mathcal{F} = L$ is a line bundle, then $L_P/\mathfrak{m}_P L_P$ is a 1-dimensional $k(P)$ -vector space, so this lemma specializes to the statement that L is globally generated at P by V iff there exists a section $f \in V$ which has $f(P) \neq 0$. When this fails, we say that P is a **base point** for V .

Proposition 2.3. Let X be quasicompact. If $\mathcal{F}$ is globally generated by $V \leq H^0(X, \mathcal{F})$ at all closed points of X , then $\mathcal{F}$ is globally generated by V at all points of X .

Proof. This is a standard specialization to closed points type argument. Let $P \in X$, and since X is quasicompact, there is a closed point specialization $\mathfrak{m} \in \overline{\{P\}}$.

Since $\mathcal{F}$ is globally generated by V at closed points, we have

$$\mathcal{O}_{X,\mathfrak{m}} \otimes_k V \twoheadrightarrow \mathcal{F}_{\mathfrak{m}},$$

and localizing at the prime ideal $P \in \mathcal{O}_{X,\mathfrak{m}}$, we get

$$\mathcal{O}_{X,P} \otimes_k V \twoheadrightarrow \mathcal{F}_P,$$

since localization is exact.

Heuristically, this proof is saying that some global sections which, doing $\mathcal{F}_{\mathfrak{m}}$ linear combinations, globally generate $\mathcal{F}$ at $\mathfrak{m}$ will also globally $\mathcal{F}$ at P (the generic point of a subscheme), where we do $\mathcal{F}_P$ linear combinations, which are much more plentiful (generating much more). $\square$

2.1.3 Embedding using a Line Bundle

Now, suppose V generates L . We show how this gives rise to a morphism from X to projective space.

Non-intrinsic way. First, we do it in a choice-y way, until we later show our choice doesn't really matter. Choose a basis $V = \langle s_0, \dots, s_r \rangle$. In analogy to our discussion about morphisms using the trivial line bundle, we will define

$$X \xrightarrow{[s_0 : \dots : s_r]} \mathbb{P}_k^r.$$

As before, this makes sense on k -points, since the tuple $(s_0(P), \dots, s_r(P)) \in (L_P/\mathfrak{m}_P L_P)^{r+1}$ can be identified (up to k^* action) with a nonzero tuple in k^{r+1} , and then we can pass to its equivalence class.

But, as before, to make full sense of this we will work locally. Since V is basepoint free (because it generates L), the open sets $D(s_i)$ cover all of X . On each $D(s_i)$, the section s_i is nonvanishing, so

$$\mathcal{O}_{D(s_i)} \xrightarrow{\cdot s_i} L|_{D(s_i)}$$

is an isomorphism (it is surjective on fibers, so on stalks, so surjective, and a surjective map of line bundles is an isomorphism).

We will abuse notation and denote the inverse of this morphism of sheaves by $s \mapsto \frac{s}{s_i}$. Now, we can define

$$D(s_i) \xrightarrow{(\frac{s_0}{s_i}, \dots, \frac{s_i}{s_i}, \dots, \frac{s_r}{s_i})} D(x_i) \subseteq \mathbb{P}_k^r.$$

One should then check that this is well-defined on all of X (that these morphisms restrict to the same morphism on pairwise intersections $D(s_i s_j)$).

Intrinsic way. Now, we show an intrinsic way that V gives a morphism $\varphi : X \rightarrow \mathbb{P}(V) = \mathbb{P}(\text{Sym}^\bullet V^*)$, the projective space of lines in V , where we send (at least for k -points)

$$P \mapsto (\text{eval}_P : V \rightarrow L/\mathfrak{m}_P L \cong k).$$

For this map $X \rightarrow \mathbb{P}(V)$ to make sense, we need eval_P to always be nonzero, and this happens exactly because V base point free, due to global generation. Also, the isomorphism $L/\mathfrak{m}_P L \cong k$ does not matter, since it would only change eval_P by scaling.

We should describe this morphism more precisely, i.e., say what happens to non k -points and specify the map of structure sheaves. To do this, we will pick coordinates and work locally. Pick a basis $V = \langle s_0, \dots, s_r \rangle$, which in turn gives a basis $V^* = \langle s_0^*, \dots, s_r^* \rangle$. Then, we observe that

$$\text{eval}_P = s_0(P)s_0^* + \dots + s_r(P)s_r^*,$$

so in fact we can describe the morphism by just

$$X \xrightarrow{[s_0 : \dots : s_r]} \text{Proj } k[s_0^*, \dots, s_r^*] \xrightarrow{\cong} \text{Proj } \text{Sym}^\bullet V^*.$$

We then call L **very ample** if φ is a closed embedding for some V . This won't necessarily be any V that generates L , for example take $X = \mathbb{P}^1$, $L = \mathcal{O}_{\mathbb{P}^1[s:t]}(2)$, and $V = \langle s^2, t^2 \rangle$. This V has

$$V \otimes \mathcal{O} \twoheadrightarrow L$$

because these functions do not simultaneously vanish, so indeed, V induces a morphism $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ by $[s : t] \mapsto [s^2 : t^2]$. However, it is *not* a closed embedding, since it's not even injective, e.g., $[-1 : 1], [1 : 1]$ map to the same point. On the other hand, if we take $V = \langle s^2, st, t^2 \rangle$, then we indeed get a closed embedding, since this is just the Veronese/2-uple embedding of $\mathbb{P}^1 \hookrightarrow \mathbb{P}^2$.

Proposition 2.4. Assume $V \leq H^0(X, L)$ generates L , so we it gives a morphism $\varphi : X \rightarrow \mathbb{P}(V)$. Then, $\varphi^*\mathcal{O}(1) = L$.

Proof. We can see this by choosing a basis $\{s_0, \dots, s_r\}$ of L so φ is defined as

$$X \xrightarrow{[s_0 : \dots : s_r]} \mathbb{P}_k^r.$$

Then, $\varphi^*\mathcal{O}(1)$ is trivial over each $\varphi^{-1}D(x_i) = D(\varphi^*x_i)$, since each has nonvanishing section φ^*x_i . In fact, $\varphi^{-1}D(x_i) = D(s_i)$. To check this, we show $\varphi^{-1}D(x_i) \cap D(s_j) = D(s_i s_j)$, so we show

$$\varphi|_{D(s_j)}^{-1}(D(x_i)) = D(s_i s_j).$$

The restricted φ is a map

$$\varphi|_{D(s_j)} : D(s_j) \rightarrow D(x_j) \cong \text{Spec } k[x_{0/j}, \dots, x_{r/j}] \hookrightarrow \mathbb{P}_k^r,$$

defined by $(\frac{s_0}{s_j}, \dots, \frac{s_r}{s_j})$, and we have as usual for pulling back functions from affine schemes, $\varphi|_{D(s_j)}^{-1}(D(x_{i/j})) = D(\frac{s_i}{s_j}) = D(s_i s_j)$ (where $D(\frac{s_i}{s_j})$ is the usual distinguished open of $\frac{s_i}{s_j}$ inside $D(s_j)$, which then includes into X as $D(s_i s_j)$).

So far, we have that both $L, \varphi^*\mathcal{O}(1)$ are trivialized by the same open cover $D(s_i)$, so all that is left is to show that they have the same transition functions. The transition function for $\varphi^*\mathcal{O}(1)$ from $D(s_i)$ to $D(s_j)$ is $\varphi^*\frac{x_i}{x_j}$, seen from chasing the trivializations

$$\mathcal{O}_{D(s_i)} \xrightarrow{\cdot x_i} \mathcal{O}(1)_{D(s_i)}, \quad \mathcal{O}(1)_{D(s_j)} \xrightarrow{\cdot \frac{1}{x_j}} \mathcal{O}_{D(s_j)}.$$

Now, $\frac{x_j}{x_i}$ is a function on $D(x_i)$, and it will pullback to a function on $D(s_i)$, where φ is locally given by the coordinates $(\frac{s_0}{s_i}, \dots, \frac{s_r}{s_i})$, so the pull back of the $\frac{x_j}{x_i}$ coordinate is $\frac{s_j}{s_i}$.

On the other hand, the transition function for L from $D(s_i)$ to $D(s_j)$ is, chasing through the trivializations

$$\mathcal{O}_{D(s_i)} \xrightarrow{\cdot s_i} L|_{D(s_i)}, \quad L|_{D(s_j)} \xrightarrow{\cdot \frac{1}{s_j}} \mathcal{O}_{D(s_j)},$$

given by $\frac{s_i}{s_j}$. Thus, we indeed have $L \cong \varphi^*\mathcal{O}(1)$. $\square$

Now, the closed embedding $X \hookrightarrow \mathbb{P}V = \text{Proj Sym}^\bullet V^*$ will correspond to some ideal sheaf $\mathcal{I}_X \leq \mathcal{O}_{\mathbb{P}V}$, and in turn

$$X \cong \text{Proj Sym}^\bullet V^* / \Gamma_*(\mathcal{I}_X),$$

which realizes X as a projective scheme. We study the following commutative diagram.

$$\begin{array}{ccc} X & \xrightarrow{\cong} & \text{Proj Sym}^\bullet V^* / \Gamma_*(\mathcal{I}_X) \\ & \searrow & \downarrow \\ & & \mathbb{P}V \end{array}$$

On one hand, the diagonal map pulls back $\mathcal{O}_{\mathbb{P}V}(1)$ to L . Then, the morphism $\text{Proj Sym}^\bullet V^* / \Gamma_*(\mathcal{I}_X) \rightarrow \mathbb{P}V$ is induced by the surjective map of graded rings

$$\text{Sym}^\bullet V^* \twoheadrightarrow \text{Sym}^\bullet V^* / \Gamma_*(\mathcal{I}_X),$$

so therefore it pulls back $\mathcal{O}_{\mathbb{P}V}(1)$ to $\mathcal{O}_{\text{Proj Sym}^\bullet V^* / \Gamma_*(\mathcal{I}_X)}(1)$. So, in total we have an isomorphism

$$X \cong \text{Proj } S,$$

which pulls back $\mathcal{O}_{\text{Proj } S}(1)$ to L , where S is a finitely generated graded ring over k generated in degree 1. This statement is in fact an alternative definition for L being very ample.

Proposition 2.5. Assume $X \cong \text{Proj } S$ for S a finitely generated graded ring over k generated in degree 1. Also assume $\mathcal{O}_{\text{Proj } S}(1)$ pulls back to L . Then, L is very ample.

Proof. Taking degree 1 generators $x_0, \dots, x_r$ for S , we get a surjection of graded rings

$$k[x_0, \dots, x_r] \twoheadrightarrow S,$$

which gives a closed embedding

$$\text{Proj } S \hookrightarrow \mathbb{P}_k^r,$$

which pulls back $\mathcal{O}_{\mathbb{P}_k^r}(1)$ to $\mathcal{O}_{\text{Proj } S}$, so the composition

$$X \xrightarrow{\cong} \text{Proj } S \rightarrow \mathbb{P}_k^r$$

pulls back $\mathcal{O}_{\mathbb{P}_k^r}(1)$ to L . If we denote the composition $\varphi : X \hookrightarrow \mathbb{P}_k^r$, then such a map is given by the section $\varphi^*x_0, \dots, \varphi^*x_r \in L$, so indeed L is very ample because it gives a map to projective space via its sections. $\square$

Remark 2.6. We could have redone this proof using the intrinsic way of embedding X in projective space by instead of surjective $k[x_0, \dots, x_r] \twoheadrightarrow S$, we could have considered $V = S_1^*$, and then considered the surjection

$$\text{Sym}^\bullet V^* \twoheadrightarrow S.$$

Next, we say a projective variety $X \hookrightarrow \mathbb{P}^n$ is **projectively normal** (equivalently $\mathcal{O}_X(1)$ is **normally generated**) if

$$H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k)) \rightarrow H^0(X, \mathcal{O}_X(k))$$

is surjective, i.e., that we don't get any "extra" sections not from restriction. We should think of projective normality as saying that resolutions of $\Gamma_*\mathcal{O}_X$ are the "same" as those of $I_X = \Gamma_*\mathcal{I}_X$. This is in the sense that when building our minimal graded free resolution of $\Gamma_*\mathcal{O}_X$, we can start by

$$S \twoheadrightarrow \mathcal{O}_X \rightarrow 0,$$

so the next step is generating the kernel, which is the same as just building a resolution of I_X . This also says that $\Gamma_*\mathcal{O}_X = S/I_X$, i.e., just the homogeneous coordinate ring of X , and again this need not be true in general.

For an example of a projective variety that is not projectively normal, take the twisted quartic, defined as follows. Take $X = \mathbb{P}^1$, $L = \mathcal{O}_{\mathbb{P}^1}_{[s:t]}(4)$, and $V = \langle s^4, s^3t, st^3, t^4 \rangle \leq H^0(L)$, so V defines

$$\begin{aligned} i : \mathbb{P}^1 &\hookrightarrow \mathbb{P}^3 \\ [s : t] &\longmapsto [s^4 : s^3t : st^3 : t^4], \end{aligned}$$

which indeed turns out to be a closed embedding. Now, we show that

$$H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \rightarrow H^0(X, \mathcal{O}_X(1))$$

is *not* surjective, by computing dimensions. On one hand, $h^0(\mathcal{O}_{\mathbb{P}^3}(1)) = 4$, while

$$\begin{aligned} h^0(\mathcal{O}_X(1)) &= h^0((i_*\mathcal{O}_{\mathbb{P}^1}) \otimes \mathcal{O}_{\mathbb{P}^3}(1)) \\ &= h^0(i_*(\mathcal{O}_{\mathbb{P}^1} \otimes i^*\mathcal{O}_{\mathbb{P}^3}(1))) && \text{(projection formula)} \\ &= h^0(i_*(\mathcal{O}_{\mathbb{P}^1} \otimes L)) \\ &= h^0(L) = 5, \end{aligned}$$

and we cannot get a surjection from $\mathbb{C}^4 \rightarrow \mathbb{C}^5$. In particular, this tells us $\Gamma_*\mathcal{O}_X \neq S/I_X$, but instead it is larger, since at least $\mathcal{O}_X(1)$ has more sections. So, instead the minimal free resolution of $E = \Gamma_*\mathcal{O}_X$ will be (you can convince yourself?? or use Macaulay2??)

$$0 \rightarrow S(-3)^3 \rightarrow S(-2)^5 \rightarrow S(-1) \oplus S \rightarrow E \rightarrow 0,$$

so while the kernel of $S \rightarrow E$ is I_X , the map $S \rightarrow E$ isn't surjective so we had to do something else, and we're going to get a bigger kernel, and the resolution will not just be that of I_X .

2.1.4 A Characterization of Very Ample

This is a bit hard to prove—we need some more technology.

Proposition 2.7. Let $k = \bar{k}$ and X a proper k -scheme. Let L be a line bundle, and $V \leq H^0(X, L)$ a k -subspace such that

- (1) for every pair of distinct closed points $x, y \in X$, there is a section in V which vanishes at x but not at y , and
- (2) for every closed point $x \in X$ and nonzero tangent vector $\theta \in T_{X/k, x}$ there exists a section $s \in V$ which vanishes at x but whose pullback by θ is nonzero.

Then, L is very ample and the morphism $\varphi : X \rightarrow \mathbb{P}(V)$ is a closed immersion.

Proof. Condition (1) in particular tells us that L is globally generated, so induces a morphism $\varphi : X \rightarrow \mathbb{P}(V)$. Because X is proper/ k , φ is proper (so in particular closed).

Even better, (1) tells us that φ is an injection on closed points, so φ will be a homeomorphism onto its image.

We now need to show

$$\varphi^\# : \mathcal{O}_{\mathbb{P}(V)} \rightarrow \varphi_* \mathcal{O}_X$$

is surjective, which we may check on closed points. Now, φ injective implies closed points have finite fibers. “A proper morphism is finite in a neighborhood of a finite fiber”, so φ is finite (since X proper). □

Corollary 2.8. Let $k = \bar{k}$, X a proper k -scheme, and L a line bundle. Assume for every closed subscheme $Z \subseteq X$ of dimension 0 and degree 2 over k , the map

$$H^0(X, L) \rightarrow H^0(Z, L|_Z)$$

is surjective. Then, L is very ample on X over k .

Proof. We cite the previous proposition. To get condition (1), we take $Z = x \cup y$, and to get (2), we take, corresponding to a tangent vector θ , Z to be the image of the closed embedding $\theta : \text{Spec } k[\varepsilon]/(\varepsilon^2) \rightarrow X$. □

2.1.5 Products

Let L_1, L_2 be basepoint free line bundles on X , giving morphisms

$$\varphi_i : X \rightarrow \mathbb{P}^{n_i},$$

so we get the product morphism $\varphi = (\varphi_1, \varphi_2) : X \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$. Now, we have the commutative diagram

$$\begin{array}{ccccc} X & \xrightarrow{\varphi} & \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} & \xrightarrow{\pi_i} & \mathbb{P}^{n_i} \\ & \searrow & \text{curved arrow} & \nearrow & \\ & & \varphi_i & & \end{array}$$

which tells us $\varphi^* \pi_i^* \mathcal{O}_{\mathbb{P}^{n_i}}(1) = L_i$. Since pullback commutes with tensors,

$$\varphi^*(\mathcal{O}_{\mathbb{P}^{n_1}} \boxtimes \mathcal{O}_{\mathbb{P}^{n_2}}) = L_1 \otimes L_2.$$

Now, the Segre embedding

$$\mathbb{P}^{n_1} \otimes \mathbb{P}^{n_2} \rightarrow \mathbb{P}^{n_1 n_2 + n_1 + n_2}$$

is defined by the complete linear series on the line bundle $\mathcal{O}_{\mathbb{P}^{n_1}}(1) \boxtimes \mathcal{O}_{\mathbb{P}^{n_2}}(1)$, so the composition

$$X \rightarrow \mathbb{P}^{n_1} \otimes \mathbb{P}^{n_2} \rightarrow \mathbb{P}^{n_1 n_2 + n_1 + n_2}$$

pulls back $\mathcal{O}_{\mathbb{P}^{n_1 n_2 + n_1 + n_2}}(1)$ to $L_1 \otimes L_2$.

2.1.6 Easy Implications of Very Ample

Proposition 2.9. Let L be very ample on X . Then, L is basepoint free. (This is only non-tautological if we take as definition $X \cong \text{Proj } S$ for S f.g. graded k -algebra generated in degree 1 with $\mathcal{O}_{\text{Proj } S}(1)$ pulls back to L).

Proof. Since sections of L gives an embedding, in particular they cannot all simultaneously vanish at any point. $\square$

Proposition 2.10. If L, B are line bundles, with L very ample and B basepoint free, then $L \otimes B$ is very ample.

Proof. Say L gives a closed embedding $X \hookrightarrow \mathbb{P}^m$, B gives a morphism $X \rightarrow \mathbb{P}^n$. Then, we recall that $L \otimes B$ gives the composite morphism

$$X \rightarrow \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^{mn+m+n},$$

where the final map is the Segre embedding, which is already a closed embedding. So, we only need to show that $X \rightarrow \mathbb{P}^m \times \mathbb{P}^n$ is a closed embedding. Since we know $X \rightarrow \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^m$ is a closed embedding, and $\mathbb{P}^m \times \mathbb{P}^n \rightarrow (\mathbb{P}^m \times \mathbb{P}^n) \times_{\mathbb{P}^m} (\mathbb{P}^m \times \mathbb{P}^n)$ (the diagonal of the projection) is a closed embedding, by Vakil's cancellation lemma, our desired map is a closed embedding. $\square$

Proposition 2.11. Let X_1, X_2 be k -schemes, and L_1, L_2 very ample. Then, $L_1 \boxtimes L_2$ on $X_1 \times X_2$ is very ample.

Proof. Suppose L_i gives a morphism $\varphi_i : X_i \rightarrow \mathbb{P}^{n_i}$. We then get a morphism

$$X_1 \times X_2 \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$$

by $\varphi = (\varphi_1 \circ \pi_1, \varphi_2 \circ \pi_2)$. By studying the diagram

$$\begin{array}{ccc} X_1 \times X_2 & \xrightarrow{\varphi} & \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \\ \downarrow \pi_i & & \downarrow p_i \\ X_i & \xrightarrow{\varphi_i} & \mathbb{P}^{n_i} \end{array}$$

we see that $\varphi^* p_i^* \mathcal{O}_{\mathbb{P}^{n_i}}(1) = \pi_i^* \varphi_i^* \mathcal{O}_{\mathbb{P}^{n_i}}(1) = \pi_i^* L_i$. So, because pull back commutes with tensors

$$\varphi^*(\mathcal{O}_{n_1}(1) \boxtimes \mathcal{O}_{n_2}(1)) = L_1 \boxtimes L_2,$$

and post composing with the Segre embedding

$$\mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \rightarrow \mathbb{P}^{n_1 n_2 + n_1 + n_2}$$

which pulls back $\mathcal{O}_{\mathbb{P}^{n_1 n_2 + n_1 + n_2}}(1)$ to $\mathcal{O}_{\mathbb{P}^{n_1}}(1) \boxtimes \mathcal{O}_{\mathbb{P}^{n_2}}(1)$, we get that $L_1 \boxtimes L_2$ defines the composite

$$X_1 \times X_2 \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2} \rightarrow \mathbb{P}^{n_1 n_2 + n_1 + n_2},$$

where the last map (the Segre embedding) is already a closed embedding. So, we only need to check that $X_1 \times X_2 \rightarrow \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$ is a closed embedding. But this is true, since closed embeddings are “reasonable” so are preserved by products. $\square$

Proposition 2.12. Let X be a k -scheme and L a very ample line bundle. Then, $L = \mathcal{O}(D)$ for some effective Cartier divisor D .

Proof. Since L is very ample, we get a closed embedding

$$\varphi : X \rightarrow \mathbb{P}_k^r$$

which pulls back $\varphi^* \mathcal{O}(1) = L$. Now, we know that $\mathcal{O}(1) \cong \mathcal{O}(H)$ for any hyperplane $H \in H^0(\mathcal{O}(1))$, since we have

TODO $\square$

2.2 Many Criteria for Ampleness

Theorem 2.13. Let X be a proper k -scheme, and L a line bundle. The following are equivalent.

- (a) For some $N > 0$, $L^{\otimes N}$ is very ample.
- (a') For all $n \gg 0$, $L^{\otimes n}$ is very ample.
- (b) For all finite type quasicoherent sheaves $\mathcal{F}$, there is n_0 such that for $n \geq n_0$, $\mathcal{F} \otimes L^{\otimes n}$ is globally generated.
- (c) As f runs over the global sections of $L^{\otimes n}$ (over all $n > 0$), the open subsets $X_f = D(f)$ form a base for the topology of X .
- (c') The open subsets X_f which are affine form a base for the topology of X .
- (c'') The open subsets X_f which are affine cover X .

In this case, we say L is **ample** (over k).

2.2.1 Proof of Criteria

Some implications are automatic, like $(a') \implies (a)$, $(c'') \implies (c') \implies (c)$.

Assuming (a) and (b) are equivalent, we prove the following, which implies $(a) \implies (a')$.

Proposition 2.14. Let X be a k -scheme, and L, M line bundles on X . Assume L is ample, then $L^{\otimes n} \otimes M$ is very ample for $n \gg 0$.

Proof. Choose n_0 such that for $n \geq n_0$, $M \otimes L^{\otimes n}$ is globally generated. Then, choose $N > 0$ such that $L^{\otimes N}$ is very ample. Now, for $n \geq n_0 + N$, we get

$$M \otimes L^{\otimes n} = (M \otimes L^{\otimes(n-N)}) \otimes L^{\otimes N}$$

is the tensor product of a globally generated line bundle and a very ample line bundle, so it is very ample. $\square$

2.2.2 Some Applications

Proposition 2.15. Every line bundle on a projective k -scheme X is the difference of two very ample line bundles.

Proof. Let M be a line bundle on X , and L a very ample line bundle. Then, there exists n such that

$$M \otimes L^{\otimes n}, \quad L^{\otimes n}$$

are both very ample, and then

$$M = (M \otimes L^{\otimes n}) \otimes L^{\otimes(-n)}$$

is a difference of two very ample line bundles. $\square$

Proposition 2.16. Let $\pi : X \rightarrow Y$ a affine morphism of proper k -schemes, and L an ample line bundle on Y . Then, π^*L is ample on X .

Proof. We use (c'') to show π^*L is ample on X . Since L is ample, there exist global sections $\{s_i\}$ where $s_i \in L^{\otimes n_i}$ such that

$$\bigcup_i Y_{s_i} = Y$$

and each Y_{s_i} is affine.

Now, the $\pi^{-1}(Y_{s_i})$ cover X , so we want to see that these are affine and complements of the vanishings of sections of $(\pi^*L)^{\otimes n} = \pi^*(L^{\otimes n})$ (since pullback commutes with tensors). Since π is finite, it is in particular affine, so $\pi^{-1}(Y_{s_i})$ is affine. Next, as always, we have

$$\pi^{-1}(Y_{s_i}) = X_{\pi^*s_i},$$

so these are complements of vanishings of sections.

So, we indeed have L is ample. $\square$

Proposition 2.17. Let L be ample on a k -scheme X , and M either ample or basepoint free. Then $L \otimes M$ is ample.

Proof. Choose $N \gg 0$ such that $M^{\otimes N}$ is basepoint free (so if M basepoint free, then $N \geq 1$, and if M is ample, then definitely very ample is enough) and such that L is very ample. Then,

$$(L \otimes M)^{\otimes N} = L^{\otimes N} \otimes M^{\otimes N}$$

is the tensor product of very ample with basepoint free, so it is very ample, so $L \otimes M$ is ample. $\square$

Proposition 2.18. Let X be a projective k -scheme, L an ample line bundle on X , M a line bundle on X , and $q_1, \dots, q_m$ points of X (not necessarily closed). Then, for all $N \gg 0$, there is a section of $L^{\otimes N} \otimes M$ that does not vanish on any of the q_i .

Proof. First, we argue that we can assume no point is a generalization of any other point. Suppose $p \in \overline{\{q\}}$ (so q is a generalization of p), and assume $s \in \Gamma(X, L^{\otimes N} \otimes M)$ is a section which vanishes in the q fiber, and we show s vanishes in p fiber (so the contrapositive says that if we can find s nonvanishing at the specialization, it doesn't vanish in the generalization).

To see this, we work in a trivialization $\text{Spec } A$ for our line bundle, where s restricts to $a \in A$, and p, q become prime ideals

$$q \leq p \leq A.$$

We know $a = 0$ in $(A/q)_q$ by assumption, and since A/q is an integral domain, the localization is injective, so $a = 0$ in A/q . Quotienting further, we get $a = 0$ in A/p , so $a = 0$ in $(A/p)_p$. Thus, s vanishes at p .

Therefore, without loss of generality, we can assume no point is a generalization of any other point.

Without Cohomology. Now, our strategy is to find a sufficiently large N such that there are sections $s_i \in \Gamma(L^{\otimes N} \otimes M)$ which vanish at all points except q_i , so that

$$s = \sum s_i$$

doesn't vanish at any of the points. We show for n bigger than some n_0 , there is a section $t = s_1$ which vanishes at $q_2, \dots, q_m$ but not q_1 . (In general, we pick n to be the max of the n_0 for the q_i s).

Let $\mathcal{I}_Z$ be the ideal sheaf of the reduced closed subscheme

$$Z = \overline{\{q_2\}} \cup \dots \cup \overline{\{q_m\}}.$$

By (b), we can choose n_0 such that for $n \geq n_0$

$$L^{\otimes n} \otimes M \otimes \mathcal{I}_Z \leq L^{\otimes n} \otimes M$$

is globally generated. Any section of the left hand sheaf, included in $L^{\otimes n} \otimes M$, vanishes in the fiber of $q_2, \dots, q_n$, since in any affine chart $\text{Spec } A$ containing q_i which trivializes $L^{\otimes n} \otimes M$, we get

$$(L^{\otimes n} \otimes M \otimes \mathcal{I}_Z)|_{\text{Spec } A} \cong \mathcal{I}_Z|_{\text{Spec } A} = \bigcap_{j=2}^m q_j \leq \tilde{A},$$

and so restricting a section t to $\text{Spec } A$, its value in the fiber (considered as a section of $L^{\otimes n} \otimes M$) will be in

$$0 = \left(\left(q_i + \bigcap_{j=2}^m q_j \right) / q_i \right)_{q_i} \leq (A/q_i A)_{q_i} \cong (L^{\otimes n} \otimes M)(q_i),$$

so indeed $t(q_i) = 0$.

On the other hand, since $q_1 \notin Z$ (this crucially uses the fact that q_1 is not a specialization of the q_i), the fiber over q_1 is $k \cong (L^{\otimes n} \otimes M)(q_1)$, since

$$(L^{\otimes n} \otimes M \otimes \mathcal{I}_Z)|_{X \setminus Z} = (L^{\otimes n} \otimes M)|_Z.$$

So, by the global generation assumption, we can pick a global section $t \in \Gamma(L^{\otimes n} \otimes M \otimes \mathcal{I})$ which generates the fiber. Now, indeed, the inclusion of t to a global section of $L^{\otimes n} \otimes M$ is a section which does not vanish at q_1 but vanishes at $q_2, \dots, q_m$.

Using Cohomology. Define Z to be the closed subscheme of $\bigcup \{\overline{q_i}\}$ and $\mathcal{I}_Z$ the ideal sheaf. Then, study the exact sequence

$$0 \rightarrow \mathcal{I}_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0,$$

which we tensor by $L^{\otimes N} \otimes M$ and take cohomology to get

$$H^0(X, L^{\otimes N} \otimes M) \rightarrow H^0(Z, (L^{\otimes N} \otimes M)|_Z) \rightarrow H^1(X, L^{\otimes N} \otimes M \otimes \mathcal{I}_Z).$$

By Serre vanishing we can get that H^1 to vanish by taking N sufficiently large. if the points were all closed points/or maybe all disjoint, then a locally free sheaf on Z would be a direct sum of $\mathcal{O}_{p_i}$ so surjecting means we can choose whichever values for each point... this argument needs work if they intersect. $\square$

Proposition 2.19. Suppose X is a projective k -scheme, M is a line bundle on X , and $q_1, \dots, q_m$ are points of X (not necessarily closed). Show that there are effective Cartier divisors D_1 and D_2 on X not containing $q_1, \dots, q_m$ such that $M \cong \mathcal{O}(D_1 - D_2)$.

Proof. Since X is projective, we can choose a very ample line bundle L , which will then correspond to an effective Cartier divisor. $\square$

3 Sheaves on Projective Space and Graded Modules

3.1 Correspondence

3.2 Resolutions

Given a finitely generated graded module E of dimension 0 over $S = \mathbb{C}[x_0, \dots, x_n]$, by the Auslander-Buchsbaum formula (since it will be finite length, so all $s \in S$ are zero divisors, so regular sequences are length 0, so depth is 0) it has a minimal graded free resolution

$$0 \rightarrow P_{n+1} \rightarrow \cdots \rightarrow P_0 \rightarrow E \rightarrow 0,$$

and the associated sheaf functor $\widetilde{(-)}$ is exact, applying it we get an exact sequence

$$0 \rightarrow \mathcal{P}_{n+1} \xrightarrow{d_{n+1}} \mathcal{P}_n \xrightarrow{d_n} \cdots \rightarrow \mathcal{P}_1 \xrightarrow{d_1} \mathcal{P}_0 \rightarrow 0, \quad (*)$$

where $\widetilde{E} = 0$ because graded dimension 0 (equivalently graded finite length) corresponds to support over the empty set. Notice that we “forgot” E .

We will study the reverse process, where we start with an exact sequence $(*)$ of sheaves on $\mathbb{P}^n$, where each $\mathcal{P}_i$ is a direct sum of twisting sheaves.

We will show that H_*^0 of this sequence is still exact, except for the last term, so there is an extra cokernel, which will be of finite length, recovering E . In particular, this will give a correspondence between finitely generated graded S -modules of dimension 0 and exact sequences of length $n + 1$ on $\mathbb{P}^n$ where matrices have entries either 0 or nonconstant.

To do this, we split up our LES into SESs as usual

$$0 \rightarrow \mathcal{P}_{n+1} \rightarrow \mathcal{P}_n \rightarrow \text{imd}_n \rightarrow 0 \quad (\mathcal{A}_n)$$

$$0 \rightarrow \text{imd}_n \rightarrow \mathcal{P}_{n-1} \rightarrow \text{imd}_{n-1} \rightarrow 0 \quad (\mathcal{A}_{n-1})$$

...

$$0 \rightarrow \text{imd}_2 \rightarrow \mathcal{P}_1 \rightarrow \mathcal{P}_0 \rightarrow 0, \quad (\mathcal{A}_1)$$

and we at least get $H_*^0(\mathcal{A}_k)$ is left exact. We need to show even more that $H_*^0(\mathcal{A}_k)$ is right exact for $k \geq 2$ (since we only are trying to show exactness everywhere *except* at $\mathcal{P}_0$).

In other words, we have exact sequences $k = 2, \dots, n - 1$ (cases $k = 2, n$ is slightly different)

$$0 \rightarrow H_*^0(\text{imd}_{k+1}) \rightarrow H_*^0(\mathcal{P}_k) \rightarrow H_*^0(\text{imd}_k) \rightarrow H_*^1(\text{imd}_{k+1}),$$

and our goal is to get vanishings of $H_*^1(\text{imd}_{k+1})$, so we can glue these together.

We go through this more carefully here. First, our goal is to show $H_*^1(\text{imd}_3) = 0$ to get the exactness of $H_*^0(\mathcal{A}_2)$. We use the LES of $H_*^\bullet$ on $\mathcal{A}_3$ giving us

$$H_*^1(\mathcal{P}_3) \rightarrow H_*^1(\text{imd}_3) \rightarrow H_*^2(\text{imd}_4),$$

and on $\mathbb{P}^n$ ($n > 1$), $H_*^1(\mathcal{P}_3) = 0$ (since it is a direct sum of $\mathcal{O}(a)$). So, it suffices to prove $H_*^2(\text{imd}_4) = 0$.

So, our new goal is to show $H_*^1(\mathrm{imd}_4), H_*^2(\mathrm{imd}_4) = 0$ (the first to get the exactness of $H_*^0(\mathcal{A}_3)$, the second exactness of $H_*^0(\mathcal{A}_2)$).

Continuing in a similar fashion, our final goal will be to show

$$H_*^1(\mathrm{imd}_n), H_*^2(\mathrm{imd}_n), \dots, H_*^{n-2}(\mathrm{imd}_n) = 0,$$

and we get this by studying the LES of $H_*^\bullet$ on $\mathcal{A}_n$, which gives

$$H_*^i(\mathcal{P}_n) \rightarrow H_*^i(\mathrm{imd}_n) \rightarrow H_*^{i+1}(\mathcal{P}_{n+1}),$$

where the left and right terms vanish for $i = 1, \dots, n-2$ as needed.

So, our conclusion is that we get an exact sequence

$$0 \rightarrow H_*^0(\mathcal{P}_{n+1}) \rightarrow H_*^0(\mathcal{P}_n) \rightarrow \dots \rightarrow H_*^0(\mathcal{P}_1) \rightarrow H_*^0(\mathcal{P}_0) \rightarrow \mathrm{coker} H_*^0(d_1) \rightarrow 0.$$

Now, if we take the associated sheaves, we get our original exact sequence, where

$$\widetilde{\mathrm{coker} H_*^0(d_1)} = \mathrm{coker} \widetilde{H_*^0(d_1)} = 0,$$

(by exactness of $\widetilde{(-)}$) which implies that in sufficiently high degrees $\mathrm{coker} H_*^0(d_1)$ is zero, so together with finite generation this gives finite length. $\square$

We can refine this argument further—we didn't need intermediate cohomology vanishing of $\mathcal{P}_1$ or $\mathcal{P}_0$, so they need not be direct sums of line bundles, any coherent sheaf will do. If we make the length of the exact sequence shorter, i.e.,

$$0 \rightarrow \mathcal{P}_r \rightarrow \dots \rightarrow \mathcal{P}_0 \rightarrow \mathcal{F} \rightarrow 0$$

for $r < n$ (so the number of maps is 1 smaller than our original setup), we can use intermediate cohomology vanishing of $\mathcal{P}_0$ to get $H_*^0(\mathcal{P}_0) \rightarrow H_*^0(\mathcal{F})$ surjective, so that applying H_*^0 we really get an exact sequence, no extra cokernel necessary.

Remark 3.1. We still have a finite length ambiguity of sorts